

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #32 – Personal Subscription & Recurring Expense Auditor

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall analyze transaction logs to identify recurring billing patterns and distinguish recurring payments from one-time payments.	High	Pass: Recurring services are correctly identified and one-time payments are not classified as subscriptions. Fail: A one-time payment is classified as a recurring subscription.	Core functionality for identifying recurring expenses.
FR-002	Functional	The system shall allow the user to view all detected subscriptions with their renewal dates and subscription amounts.	High	Pass: Each detected subscription displays its name, amount, and next renewal date. Fail: Any detected subscription is missing one of these details.	Helps users track upcoming subscription charges.
FR-003	Functional	The system shall alert the user before a subscription reaches its renewal date.	High	Pass: An alert is generated before the configured renewal date. Fail: No alert is generated before renewal.	Prevents unexpected recurring charges.
FR-004	Functional	The system shall calculate and display the user's total monthly recurring subscription expenditure.	High	Pass: The displayed monthly total equals the sum of all active recurring subscriptions. Fail: The calculated total differs from the subscription amounts.	Helps users understand recurring spending.
FR-005	Functional	The system	Medium	Pass: Selecting	Helps users

		shall provide cancellation guidance for a selected subscription.		a subscription displays available cancellation instructions. Fail: No cancellation guidance is displayed.	manage unwanted subscriptions.
NFR-001	Nonfunctional	Financial transaction imports (CSV/OFX) must parse securely without storing raw banking credentials in cleartext.	High	Pass: Security testing confirms credentials are not stored in cleartext. Fail: Any raw banking credential is stored in cleartext.	Protects sensitive financial information.
NFR-002	Nonfunctional	The system shall display subscription analysis results within 3 seconds after valid transaction data is uploaded.	Medium	Pass: Results are displayed within 3 seconds in at least 95 out of 100 test runs. Fail: The target is not achieved.	Provides a responsive user experience.

Actors: Individual User, Finance Auditor, Bank Transaction Source

Use Cases: Import Transaction Data; Analyze Recurring Transactions; View Subscription Details; Calculate Monthly Subscription Expenditure; Receive Renewal Alert; Get Cancellation Guidance; Review Expense Report

Required relationships: Import Transaction Data <<include>> Analyze Recurring Transactions; Get Cancellation Guidance <<extend>> View Subscription Details.